Project 1: Wine Quality - Polynomial Regression and Model Selection

Student: Karan Rooprai Student ID: N12498122

First the missing values and duplicate rows were cleaned, leaving us with 4421 samples. Subsequently, the numeric chemical features were standardised with *StandardScaler*. The data split used is 60/20/20 for training, validation and testing set using *random_state=42*. For every degree '*d = 1 to 6*', *PolynomialFeatures(d)* expanded the training features and an ordinary *LinearRegression* was fitted and *mean_squared_error* (MSE) was recorded for the training and validation sets. Post that the degree with the lowest validation MSE was rebuilt as a single *make_pipeline,* refitted and then saved as *best_model.joblib.*

Degree	Training MSE	Validation MSE
1	0.5361	0.5716
2	**0.4791**	**0.5476**
3	0.4075	0.8394
4	0.2074	299.2788
5	0.0639	114,296.3322
6	0.2031	6,829.9272

Table 1. Mean Squared Error for Training and Validation set

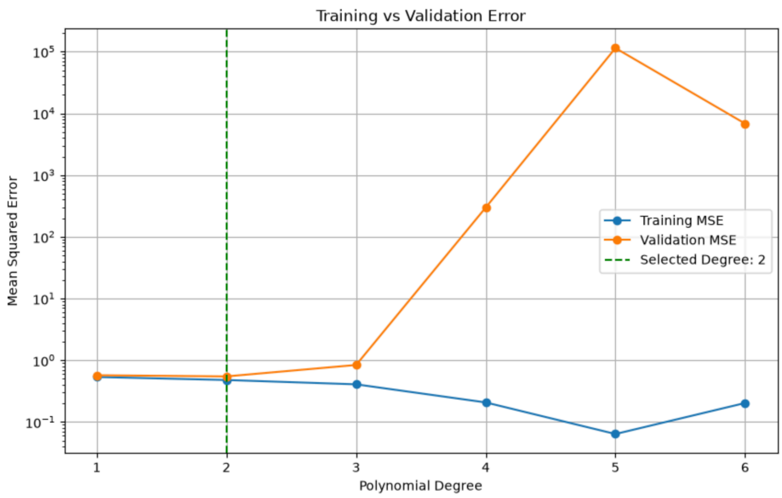

Figure 1. Training and Validation MSE against polynomial degree

A. Trends observed as the model becomes more complex

From Figure 1, we can observe that the two lines separate sharply. The training MSE drops steadily from degree 1 (*0.5361*) to degree 5 (*0.0639*). While the degree increases, the function space enlarges as well. Whereas Validation MSE is the opposite, it improves once from degree 1 (*0.5716*) to degree 2 (*0.5476*), which is 4.2%, then increases significantly from degree 3 to 5. Here, a validation MSE in the hundreds is meaningless, so every degree after 3 is unfit for use.

Two features are not explained by bias-variance trade-off alone. Training MSE rises from degree 5 (*0.0639*) to degree 6 (*0.2031*), meanwhile the validation MSE falls from *114296.33* to *6829.93* at the same step. These reversals seem to be a trait of computation rather than the model itself. By degree 5 the expansion produces thousands of highly correlated terms from 11 features which leaves the matrix ill-conditioned.

B. Where training and validation set performance decouple

The model's performance on the training set starts to decouple from its performance on the validation set at degree 3. Up to degree 2 the line falls together and stay close. Validation error exceeds training error by only *0.0355* at degree 1 and *0.0685* at degree 2. At degree 3, training error continues to go down to *0.4075* while validation error starts to increase to *0.8394*, making the gap significant.

C. Diagnosis

Low Degrees - High Bias and Low Variance.

At degree 1, the training and validation errors are both high and almost similar i.e. *0.5361* and *0.5716.* This type of similar size error on seen and unseen data is a sign of underfitting. A strictly linear model is too rigid to capture the curve and interactions between chemical properties. Whereas at degree 2, we can observe that the bias reduces and variance does not rise significantly and they both improve.

High Degrees - Low Bias and High Variance.

At degree 5, the training MSE is *0.0639* which indicates that the model is reproducing every training label and the validation MSE of *114296.33* corresponds to predictions wrong by hundreds of points. It indicates that it has fitted the noise and precise positions of training wines rather than finding any meaningful relationship, which means any different sample would produce entirely different coefficients.

D. Recommended degree for the production model

The recommended polynomial degree for the production model is Degree 2. It is the only degree that outperforms the other degrees on unseen data. With a validation MSE of *0.5476*, it is slightly better than degree 1 (*0.5716*) and outperforms the others. The saved pipeline achieves MSE *0.5009,* RMSE *0.7078,* MAE *0.5510* and R2 score of *0.3275,* which is a 33% reduction in MSE against the baseline. Validation and test errors differ by only 0.0467 which confirms that at this degree the model did not overfit.

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Test MSE: 0.5009
Test RMSE: 0.7078
Test MAE: 0.5510
Test R2: 0.3275
Baseline MSE: 0.7474
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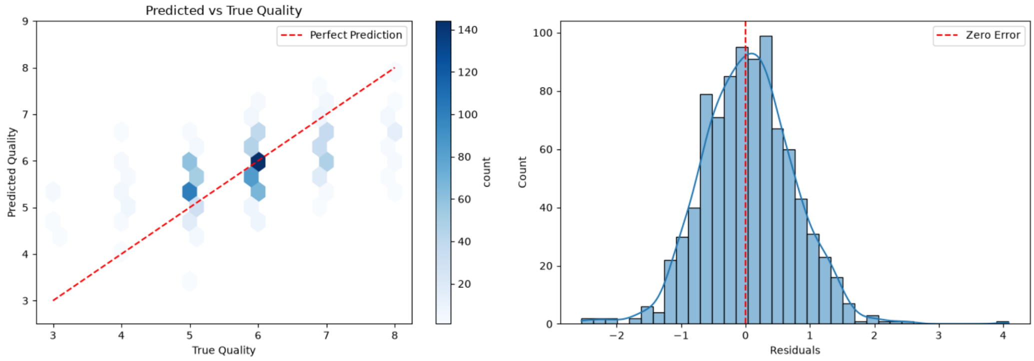

Figure 2. Left: Predicted vs True Quality. Right: Distributions of Residuals

Limitations and Recommendations

The recommended degree 2 model has limits. Its R2 is *0.3275* which means it explains only about a third of the variation in wine quality. To explicate, most of what determines a wine's quality score comes from factors that the current 11 chemical measurements might not capture. It also tends to predict close to the average as seen in Figure 2. The degree was also chosen from a single validation split. As one split can happen to favour a particular degree, repeating the selection with K-fold cross-validation would give a more reliable choice. It averages the validation error across several folds rather than trusting a single split. Moreover, collecting features beyond just 11 chemical properties could also prove to be meaningful to raise the model's explanatory power.